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ENERGY
Group 10

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Contents

1 HJM–NIG model and admissible parameters

We consider a Heath–Jarrow–Morton (HJM) framework for electricity forward prices, where the source of randomness is a Normal Inverse Gaussian (NIG) Lévy process, following the specification of Benth (2008) with parameters $p = 0$ and $n = 1$. This approach is particularly suitable for electricity markets, where forward prices are modeled directly due to the non-storability of power.

Let $F(t, T_1, T_2)$ denote the forward price at time t for the delivery period $[T_1, T_2]$. The forward dynamics are assumed to be of exponential form:

$$F(t, T_1, T_2) = F(t_0, T_1, T_2) \exp\left(\int_{t_0}^{t} a(u, T_1, T_2) du + \int_{t_0}^{t} \Upsilon(u, T_1, T_2) dL_u \right), \tag{1}$$

where L_t is a NIG Lévy process and $\Upsilon = \Upsilon(t, T_1, T_2)$ is assumed to be constant. The deterministic term

$$A(t) := \int_{t_0}^{t} a(u, T_1, T_2)\, du \tag{2}$$

is introduced to ensure the martingale property of the forward price.

The admissible parameter set is given by

$$\eta \in \mathbb{R}, \qquad \sigma > 0, \qquad k > 0, \qquad \Upsilon \in \mathbb{R} \setminus \{0\}.$$

This choice guarantees that the driving Normal Inverse Gaussian (NIG) Lévy process is well defined. The parameter σ represents the average level of volatility and must be strictly positive to avoid degeneracy of the diffusion component. The parameter k controls the volatility of volatility and is also required to be positive to ensure finite second moments of the stochastic integral. The parameter η governs the skewness of the NIG distribution and may take any real value.

The process Υ, appearing in the Benth (2008) specification, is assumed to be a real-valued continuous function and, in our framework, is taken to be constant.

For the good definition of the HJM model, one should in principle verify the assumptions of **Proposition 6.3** in Benth (2008), which provide sufficient conditions for the stochastic integral to be well defined. However, thanks to the assumption that Υ is constant, the associated Lévy measure has finite mass away from zero. This guarantees the integrability of the jump component and ensures that the resulting forward prices are well-defined martingales, together with the requirement that the exponential moment

$$\mathbb{E}\big[e^{\Upsilon L_t}\big]$$

exists. This condition guarantees that the forward price process is well-defined and finite.

Proposition 6.3 Suppose for each $j = 1, \ldots, n$, that the exponential integrability condition

$$\int_0^{\tau_1} \int_{|z|\geq 1} \exp\big(\Upsilon_j(u, \tau_1, \tau_2) z\big)\, \nu_j(dz, du) < \infty$$

holds for all $0 \leq \tau_1 \leq \tau_2 \leq T$.

2 Martingale condition and drift adjustment

We model the forward price as martingale process under the risk neutral measure, that is

$$\mathbb{E}[F(t, T_1, T_2)] = F(0, T_1, T_2). \tag{3}$$

Substituting the exponential representation of the forward price, the martingale condition becomes

$$\mathbb{E}[\exp(A(t) + \Upsilon L_t)] = 1. \tag{4}$$

Let $\phi_t(u)$ denote the characteristic function of the NIG Lévy process L_t
Therefore, the martingale condition can be written as

$$e^{A(t)}\, \phi_t(-i\Upsilon) = 1, \tag{5}$$

which yields the explicit expression for the drift adjustment:

$$\boxed{A(t) = -\log(\phi_t(-i\Upsilon))\,.} \tag{6}$$

For a NIG Lévy process, as we saw in class the characteristic function is given by

$$\phi_t(u) = \exp\left(L_t\left(\frac{u^2}{2} + \frac{iu}{2} + iu\eta\right)\right), \tag{7}$$

where the Laplace exponent $L_t(s)$ reads

$$L_t(s) = \frac{t}{k}\left(1 - \sqrt{1 + 2k\sigma^2 s}\right). \tag{8}$$

Finally we get the final formula for the risk neutral drift:

$$\boxed{A(t) = -\frac{t}{k}\left[1 - \sqrt{1 - k\sigma^2\Upsilon^2 + k\sigma^2\Upsilon(1+2\eta)}\right]}$$

This explicit formulation allows the drift term to be computed analytically, ensuring the martingale property of the forward price without relying on numerical integration.

3 Calibration on the 2027 French options

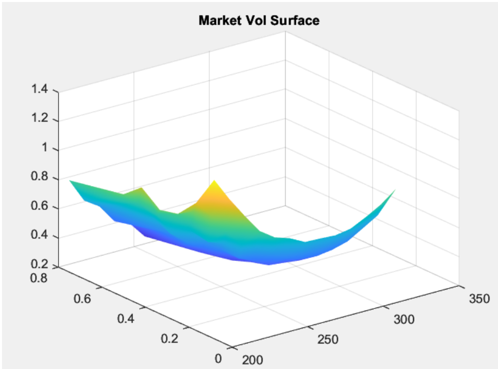
Using the given data we calibrated our model minimizing the least-squared distance between the market prices of the calls, obtained with the Black formula, and the model prices, obtained with the Lewis formula containing the four parameters of the NIG model, unfortunately the first fit obtained was terrible, so we tried to apply some filters.

3.1 Filtered dataset

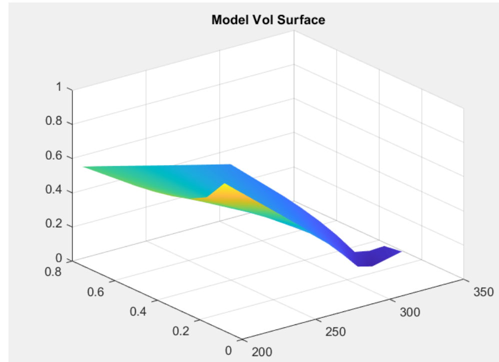
We decided to apply the following filters:

- **Penny Option Filter**: We avoided the options that had a very small price since they are one of the first causes of a poor fit leading to negative prices.
- **Delta Filter**: We decided to include only options with a Delta between 25% and 75%, excluding options that were deeply In or Out of the money.
- **Monotony Filter**: Some of the given volatilities were excluded since they were so high that caused an increasing price with higher strikes and same time to maturity, which is unreasonable.

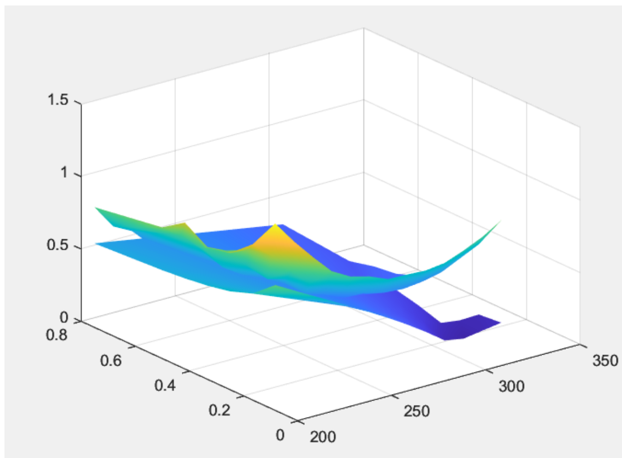
σ	η	K	Υ
1.9788	0.4866	3.8322	0.1489



(a) Market implied smile



(b) Model implied smile

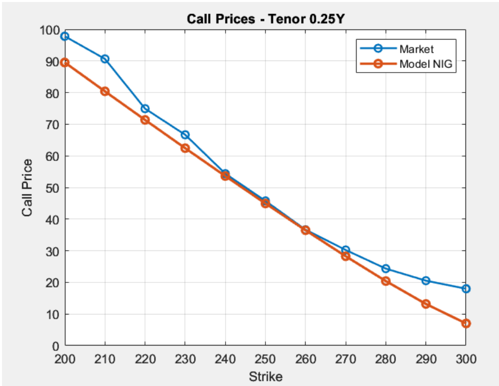


(c) Volatility smile comparison

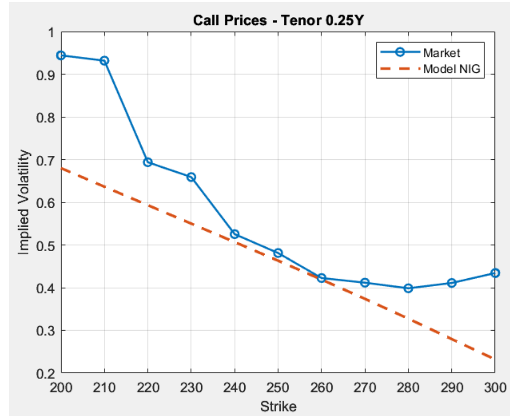
Figure 1: Volatility smile analysis

3.2 Criticalities of the Calibration procedure

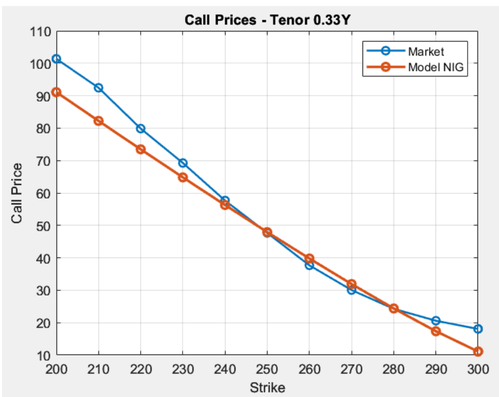
Despite the filtering procedure the results were not satisfying at all (MAPE=30.36% and RMSE=0.2484 in the fitted region), generating some concerns about the ability of the NIG model to capture the shape of the volatility surface, probably depending on the illiquidity of this specific Energy Market.
Indeed as we can see on the following graphs, in some tenors the model fits well enough the market prices, but still generates not negligible errors when we look at the corresponding volatility smiles.



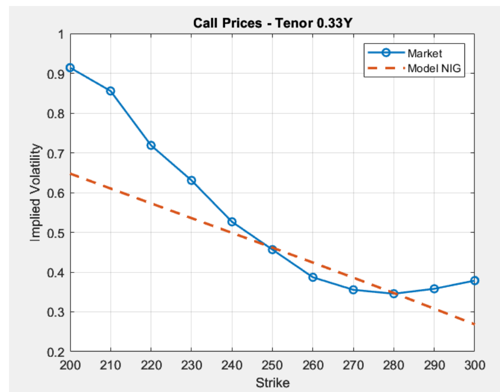
(a) Calibrated Call Prices - Tenor 0.25Y



(b) Implied Volatilities - Tenor 0.25Y



(c) Calibrated Call Prices - Tenor 0.33Y



(d) Implied Volatilities - Tenor 0.33Y

Figure 2: Volatility smile analysis

4 Pricing of a lookback option on a forward

We consider an option with payoff at time $t = 1$ given by

$$\left(\max_{t \in (0,1)} F(t; T_1, T_2) - K \right)^+,$$

where T_1 is January 1, 2027, T_2 is December 31, 2027, and $K = 300$.

4.1 Existence of a closed formula

The payoff under consideration is that of a continuously monitored lookback option written on a forward contract. More precisely, it depends on the running maximum of the forward price process $F(t; T_1, T_2)$ over the whole time interval $t \in (0, 1)$. This path–dependence already represents a major obstacle to the derivation of a closed–form pricing formula.

Indeed, we did not find any explicit pricing formula for this option. The absence of a closed form is therefore not merely a technical limitation, but a structural feature of the model and of the payoff itself. This motivates the use of numerical methods, and in particular Monte Carlo simulation, as the natural and most flexible approach for pricing such path-dependent derivatives.

4.2 Numerical pricing approach

Since no closed-form solution is available for the lookback option under consideration, the pricing must be carried out using a numerical method. We adopt a Monte Carlo approach, which is well suited for path-dependent payoffs and models driven by Lévy processes.
The main difficulty arises from the fact that the payoff depends on the continuous monitoring of the forward price over the time interval $(0, 1)$. In practice, continuous monitoring is approximated by discretizing the time interval into a sufficiently fine grid

$$0 = t_0 < t_1 < \cdots < t_N = 1,$$

and monitoring the forward price at each discretization point. The maximum of the forward price is then approximated by the maximum over this discrete set of observations.
Once the model parameters have been calibrated, simulation is performed using the known characteristic function of the log-forward price, which is derived explicitly in Section 2 of this report. Exploiting the additivity of log-returns of the forward price, the characteristic function of each increment can be obtained as the ratio of the characteristic functions at consecutive time points. This property allows us to simulate the increments efficiently and consistently with the assumed HJM–NIG dynamics.
All simulated increments are stored in a matrix, where each row corresponds to a single Monte Carlo trajectory. By applying a cumulative sum along the time dimension, we recover the full simulated paths of the log-forward price, and hence of the forward price itself. For each trajectory, we then compute the maximum value attained over the monitoring grid and evaluate the corresponding payoff

$$\left(\max_{t \in (0,1)} F(t; T_1, T_2) - K \right)^+ .$$

Finally, the option price is obtained by averaging the discounted payoffs over all simulated trajectories. Using this Monte Carlo procedure, we obtain:

$$\text{Price} = \mathbf{11.91}$$

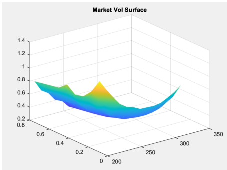
Due to poor model calibration and incorrect parameters, the price is significantly lower than expected.

5 Calibration on the 2027 and 2029 French options

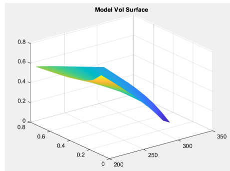
Following the same procedure as the previous calibration and maintaining the same filters, the results were still not satisfactory.

σ	η	K	Υ
1.9726	0.3279	4.0595	0.1977

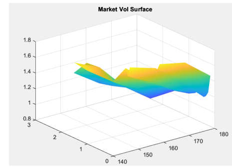
The parameters of the second calibration remain very close to the first because the 2029 surface exhibits significantly higher volatility. Our filtering strategy consequently discards more data points compared to the initial run. As a result, the minimization weight is more heavily distributed toward fitting the prices of the earlier dates. This has led to a severe misalignment between model and market prices across the 2029 volatility surface.



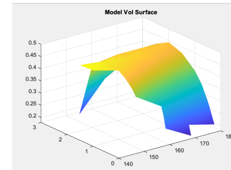
(a) Market smile - 2027



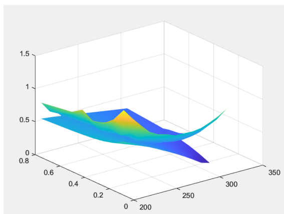
(b) Model smile - 2027



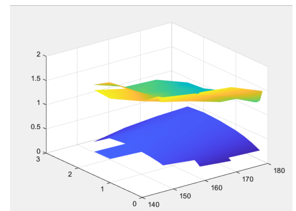
(e) Market smile - 2029



(f) Model smile - 2029



(c) Smile comparison - 2027



(g) Smile comparison - 2029

(d) Volatility smile analysis for 2027

(h) Volatility smile analysis for 2029

5.1 Criticalities of the Calibration procedure

As expected the illiquidity of the market is still a concern in the results obtained alongside the obvious decrease in precision due to the larger number of call prices to fit maintaining the same model NIG with four parameters. As goodness indicators for our fitting procedure we used the Mean Absolute Percentage Error and the Mean Rooted Squared Error.

- In the 2027 region we obtained a MAPE equal to 29.8715% and a MRSE equal to 0.2163.
- In the 2029 region we obtained a MAPE equal to 72.91% and a MRSE equal to 1.0643.

6 Pricing an option on the maximum of two forwards

Using the model introduced in point (v), we consider an option paying at time $t = 1$ with payoff

$$\left(\max\bigl(F(1;T_1,T_2),\, F(1;T_3,T_4)\bigr) - K\right)^+,$$

where T_3 is January 1, 2029 and T_4 is December 31, 2029. We investigate whether an explicit pricing formula exists or whether Monte Carlo methods are required.

As a first step, we investigate whether an explicit pricing formula can be derived for this payoff. Unlike the lookback option considered previously, the payoff depends only on the values of the two forward prices at the single maturity date $t = 1$, and therefore no continuous monitoring of the trajectories is required. This naturally raises the question of whether an analytical solution may exist.

The main obstacle we encounter is the lack of an explicit and tractable joint distribution for the two forward prices

$$F(1;T_1,T_2) \quad \text{and} \quad F(1;T_3,T_4).$$

In particular, the pricing problem crucially depends on the correlation structure between the two processes. From an economic perspective, one would expect a high positive correlation, since both forwards are written on the same underlying market and differ only by their delivery periods. However, this correlation is not perfect in general.

If perfect correlation were to hold, the maximum of the two forwards at time $t = 1$ would simply coincide with the larger of the two forward prices observed today, since Y does not depend on the delivery period. In this special case, the option payoff would reduce to that of a standard European call option on a single forward, and an explicit pricing formula would be available via the Lewis Fourier-based representation for exponential Lévy models, in particular for the NIG specification.

On the other hand, assuming independence between the two forward processes—which is a much less realistic assumption from a modeling standpoint—does not lead to an explicit pricing formula either. Indeed, the distribution of the maximum of two independent random variables depends on both the density and the cumulative distribution functions of the underlying laws. In the present case, the cumulative distribution function of the NIG distribution is not available in closed form. Moreover, the characteristic function of the maximum of two random variables does not admit a simple analytical expression, even under independence. Given these limitations, we conclude that no explicit pricing formula is available in a realistic setting with imperfect correlation, nor under the simplifying assumption of independence. As a practical workaround, we therefore opt to assume perfect correlation between the two forward prices. Under this assumption, the problem reduces to the pricing of a European call option within the exponential Lévy NIG framework, which can be evaluated efficiently using the Lewis formula.

$$\text{Price} = \mathbf{39.83}$$